DATA SCIENCE AND VISUALIZATION

Project Planning and Dataset Selection

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team Number: Team 9

1. INTRODUCTION

The first week focused on team formation, dataset selection, defining the problem statement and objectives, and planning the project workflow. The team selected a Spotify tracks dataset containing information about songs, artists, albums, popularity, and audio features.

The team also learned about Pandas and NumPy for data analysis and identified the tools required for the project. These activities established the foundation for exploring the relationship between musical characteristics and song popularity in the upcoming stages.

2. PROJECT TITLE

Discovering the Factors Behind Song Popularity Through Data Science

3. PROBLEM STATEMENT

Music streaming platforms contain a large number of songs with different musical characteristics and popularity levels. Understanding the characteristics associated with song popularity can help reveal meaningful patterns in music data.

This project aims to analyze Spotify track data using data science and visualization techniques. The project will involve understanding the dataset, assessing its quality, handling missing values and inconsistencies, performing exploratory data analysis, and investigating relationships between audio features and popularity.

The findings will be communicated through suitable visualizations and, in the later stages, an interactive dashboard.

4. OBJECTIVES

The main objectives of the project are:

- To select and understand the Spotify dataset and its attributes.
- To learn Pandas and NumPy for data analysis.
- To identify data-quality issues and perform exploratory data analysis.
- To analyze relationships between musical features and popularity, create visualizations, and prepare data for advanced analysis and dashboard development.

5. TEAM FORMATION

The project team consists of four members who will collaboratively work on dataset preparation, analysis, visualization, documentation, and presentation.

S. No.	Team Member	Register Number
1	Prithesa R	25BAD083
2	Saravanan A	25BAD102
3	Tejas T	25BAD124
4	Viknesh T	25BAD128

The team will share responsibilities to ensure that each stage of the project is completed systematically. Specific individual task assignments can be documented once finalized by the team.

6. DATASET SELECTION

6.1 Dataset Description

The team selected the Spotify tracks dataset, containing 19,335 records and 25 columns covering song, artist, album, popularity, and audio characteristics.

6.2 Important Attributes

The dataset contains the following categories of information:

- **Track information:** Track ID, track name, track number, and album details.
- **Artist information:** Artist names associated with tracks.
- **Album information:** Album name, album type, release date, and number of tracks in an album.

- **Popularity information:** Popularity values associated with tracks.
- **Audio features:** Danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.
- **Musical attributes:** Key, mode, and time signature.

These attributes provide a combination of numerical, categorical, and textual data that can be examined using statistical and visualization techniques.

6.3 Reasons for Dataset Selection

The Spotify dataset was selected for the following reasons:

1. Represents real-world music data with details about songs, artists, and popularity.
2. Contains multiple attributes and audio features suitable for statistical and comparative analysis.
3. Provides opportunities to practise data cleaning, preprocessing, exploratory data analysis, correlation analysis, and visualization.
4. Supports advanced techniques such as feature engineering and Principal Component Analysis (PCA).

7. INTRODUCTION TO PANDAS

7.1 Definition

Pandas is an open-source Python library used for data manipulation and analysis. It provides two important data structures: Series and DataFrame.

A Series represents one-dimensional labelled data, whereas a DataFrame represents two-dimensional tabular data arranged in rows and columns.

7.2 Applications of Pandas

Pandas provides several useful operations for working with datasets:

1. Loading datasets from CSV and other file formats and displaying the first and last few records.
2. Examining the number of rows and columns, column names, and data types.
3. Identifying missing values and duplicate records and generating descriptive statistics.
4. Filtering, sorting, and transforming data for further analysis.

7.3 Application in Our Project

Pandas will be used to load and inspect the Spotify dataset. It will help the team understand the dataset structure, identify data-quality issues, and prepare the data for cleaning and exploratory analysis.

Example:

```
import pandas as pd
df = pd.read_csv("DSV_TEAM-9_RAW_DATASET.csv")
print(df.head())
print(df.shape)
```

The read_csv() function loads the dataset, head() displays the first five rows by default, shape returns the number of rows and columns.

8. INTRODUCTION TO NUMPY

8.1 Definition

NumPy, which stands for Numerical Python, is a Python library used for numerical computing. It provides the multidimensional array structure and mathematical functions required for efficient numerical operations.

8.2 Applications of NumPy

NumPy supports:

- Creating and manipulating numerical arrays for data processing.
- Performing mathematical and statistical calculations, including mean, median, and standard deviation.
- Carrying out element-wise operations and numerical transformations.
- Supporting data preprocessing, statistical analysis, and future PCA workflows.

8.3 Application in Our Project

NumPy will be used to perform numerical calculations on relevant Spotify audio features. It can help calculate statistical measures and prepare numerical data for further analysis.

Example:

```
import numpy as np
audio_features = np.array([0.75, 0.82, 0.68, 0.90, 0.73])
```

```
print("Average Audio Feature:", np.mean(audio_features))
print("Standard Deviation:", np.std(audio_features))
```

The `array()` function creates a numerical array, `mean()` calculates its average, and `std()` calculates its standard deviation.

9. PROPOSED PROJECT METHODOLOGY

The project will follow a systematic data science workflow to study Spotify song popularity and explore relationships between musical characteristics and popularity.

The planned workflow is:

1. **Raw Dataset Selection:** Select the Spotify tracks dataset for the project.
2. **Data Understanding:** Study the dataset structure, attributes, and meaning of the available information.
3. **Data Cleaning:** Plan the preparation of the dataset for reliable analysis.
4. **Exploratory Data Analysis:** Explore patterns and relationships among relevant variables.
5. **Feature Engineering:** Prepare and transform suitable attributes for analysis.
6. **Feature Selection:** Identify relevant features for investigating song popularity.
7. **Principal Component Analysis (PCA):** Explore dimensionality reduction where appropriate.
8. **Data Visualization:** Develop charts and visual representations of the findings.
9. **Insights and Data Storytelling:** Present the major findings in an understandable form.

12. TOOLS AND TECHNOLOGIES

The following tools and technologies are planned for the project:

- **Python:** Primary programming language for data analysis.
- **Pandas:** Data manipulation and handling tabular datasets.
- **NumPy:** Numerical calculations and mathematical operations.
- **Matplotlib:** Creating graphs and charts.
- **Seaborn:** Statistical data visualization.
- **Scikit-learn:** Supporting feature processing and PCA.
- **Jupyter Notebook:** Writing, executing, and documenting Python code.
- **GitHub:** Version control and team collaboration.
- **Streamlit or Plotly:** Potential tools for developing an interactive dashboard.

These tools will support the different stages of the project, from data preparation to visualization and presentation.

13. WEEK 1 ACTIVITIES AND MILESTONES

The main activities planned and undertaken during Week 1 were:

- Formed the project team, selected the Spotify tracks dataset, and finalized the project title.
- Defined the problem statement, project objectives, and overall project scope.
- Introduced Pandas and NumPy and identified the required software tools and libraries.
- Planned the project workflow, established GitHub for collaboration, and prepared the initial documentation.

These activities established the foundation for the subsequent stages of the project.

14. EXPECTED OUTCOMES

The expected outcomes of the project are:

- Understand the Spotify dataset and its attributes.
- Analyze the relationship between audio features and song popularity.
- Create visualizations to identify trends and important features.
- Develop an interactive dashboard to present the findings and insights.

15. CONCLUSION

Week 1 established the foundation for the Data Science and Visualization project titled *Discovering the Factors Behind Song Popularity Through Data Science*.

The team selected the Spotify tracks dataset, defined the problem statement and objectives, and planned the overall project methodology. The team also introduced Pandas and NumPy as essential Python libraries for data manipulation and numerical computation. GitHub was identified as the platform for maintaining project files and supporting collaboration.

These initial activities provide a clear direction for the project. In the following weeks, the team will proceed with the scheduled stages of dataset understanding, preparation, analysis, and visualization to investigate the factors associated with Spotify song popularity.

DATA SCIENCE AND VISUALIZATION-WEEK 2 REPORT

Know Your Data

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team: 9

1. Introduction

Data understanding is an essential stage in the data science workflow. Before performing data cleaning, statistical analysis, or visualization, it is necessary to examine the structure, characteristics, and quality of the dataset.

During Week 2, the Spotify Songs dataset was explored using Python, Pandas, and NumPy. The analysis focused on inspecting the dataset, understanding its data types, calculating preliminary descriptive statistics, identifying missing values and duplicate records, and generating an automated First Impression Data Quality Report.

2. Aim

To load, inspect, and understand the structure and quality of the Spotify Songs dataset using Python, Pandas, and NumPy.

3. Objectives

1. Load the Spotify Songs dataset from a CSV file using Pandas.
2. Examine the dataset dimensions, first and last five records, column names, and data types.
3. Calculate preliminary descriptive statistics for numerical and categorical attributes.
4. Identify missing values, duplicate records, and potential data quality issues.
5. Generate an automated First Impression Data Quality Report and export the results as CSV files.

4. Tools and Technologies Used

- **Python:** Used for dataset exploration and analysis.
- **Pandas:** Used to load, inspect, and summarize the dataset.
- **NumPy:** Used for numerical operations.
- **Jupyter Notebook:** Used to execute the code and document the results.
- **CSV:** Used for storing the original dataset and exported reports.

5. Dataset Overview

The Spotify Songs dataset was obtained from Kaggle and contains information about songs, artists, albums, popularity, and audio characteristics.

The dataset includes attributes such as track name, artist information, album details, popularity, danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

Dataset details:

Attribute	Description
Total records	19,335
Total columns	46
File format	CSV
Data types	String, float, and integer
Memory usage	Approximately 74.65 MB

Each row represents a song record, while the columns contain information describing its metadata and audio features.

6. Dataset Loading and Inspection

The dataset was loaded into a Pandas DataFrame using `pd.read_csv()`. The `head()` and `tail()` functions were used to display the first five and last five records.

The `shape`, `columns`, `dtypes`, and `info()` functions were used to examine the dataset structure, column names, data types, non-null counts, and memory usage.

Observation: The dataset contains 19,335 rows and 46 columns. It includes both numerical audio features and textual metadata. Some attributes that represent numerical information are stored as strings and may require further validation.

7. Preliminary Descriptive Statistics

The `describe()` function was used to obtain an initial statistical summary of numerical and categorical attributes.

For numerical columns, the summary included count, mean, standard deviation, minimum, quartiles, and maximum. For categorical columns, it included the number of non-null entries, unique values, the most frequent value, and its frequency.

The numerical summary provided an initial understanding of the audio features. For example, the mean danceability was approximately **0.616**, mean energy was **0.584**, and mean tempo was approximately **119.79 BPM**.

Observation: The summaries provided an overview of the dataset's values and distributions. Detailed statistical analysis was left for a later stage of the project.

8. Missing Value Analysis

Missing values were identified using the `isnull()` and `sum()` functions. Missing percentages were calculated to evaluate the completeness of individual columns.

The analysis produced the following results:

Metric	Result
Total missing cells	7,707
Rows containing missing values	7,532
Columns containing missing values	39
Missing values in `preview_url`	7,524
Missing percentage in `preview_url`	38.91%

The `preview_url` column contained the largest number of missing values. Several audio feature and metadata columns also contained missing values, although some had only seven missing entries each.

Observation: Missing values were identified and documented without modifying the dataset. Appropriate handling decisions were reserved for the data cleaning stage.

9. Duplicate Record Analysis

The `duplicated()` function was used to identify exact duplicate rows. The Spotify song identifier column, `id`, was also checked separately for repeated identifiers.

The results were:

- **Exact duplicate rows:** 0
- **Repeated Spotify IDs:** 2

Observation: No exact duplicate rows were found. However, two repeated Spotify IDs were identified and require further investigation before deciding whether any records should be removed.

No records were deleted during Week 2.

10. Data Type Analysis

The data types of all columns were examined using `dtypes` and a data type summary table.

The dataset contained:

Data type	Number of columns
String (`str`)	32
Float (`float64`)	13
Integer (`int64`)	1
Total	**46**

The analysis also checked whether selected fields containing numerical information could be converted into numeric values.

Column	Invalid numerical values
`popularity`	1
`track_number`	1
`duration_ms.x`	4

Observation: Some numerical fields were stored as strings, and a small number of values could not be converted directly into numbers. These inconsistencies were documented for further examination. No data type corrections were made during Week 2.

11. Automated First Impression Data Quality Report

An automated First Impression Data Quality Report was generated using Pandas. It summarized the condition of all 46 columns and included:

- Column names and data types.
- Non-null counts.
- Missing value counts and percentages.
- Number of unique values.

A separate dataset summary was also generated to record the total rows, total columns, missing cells, rows containing missing values, columns containing missing values, exact duplicate rows, and memory usage.

The following CSV reports were exported successfully:

1. `Week2_Data_Quality_Report.csv`
2. `Week2_Dataset_Summary.csv`

Observation: The reports provide a structured overview of the dataset's completeness and quality. They will serve as references for planning the next stage of preprocessing.

12. Team Member Participation

The following is a proposed allocation of responsibilities based on the activities performed during Week 2.

S. No.	Team Member	Roll Number	Contribution
1	Prithesa R	25BAD083	Dataset loading, initial inspection, and understanding the dataset structure.
2	Saravanan A	25BAD102	Preliminary descriptive statistics and data type analysis.
3	Tejas T	25BAD124	Missing value analysis and duplicate record verification.
4	Viknesh T	25BAD128	Automated data quality report generation and report documentation.

13. Challenges and Considerations

1. The dataset contains missing values across multiple columns, particularly `preview_url`.
2. Some numerical attributes are stored as strings and require validation.
3. Repeated Spotify IDs need investigation even though no exact duplicate rows were found.
4. Metadata and URL fields require consideration when preparing the dataset for further analysis.

These findings were documented without changing the original dataset.

14. Conclusion

During Week 2, the Spotify Songs dataset was successfully loaded and examined using Python, Pandas, and NumPy. Its structure, data types, preliminary descriptive statistics, missing values, and duplicate records were analyzed.

The investigation identified 7,707 missing cells, no exact duplicate rows, two repeated Spotify IDs, and several values that require numerical validation. An automated First Impression Data Quality Report and a dataset summary were generated and exported successfully.

The original dataset was preserved without modification. The findings establish a foundation for the data cleaning and preprocessing activities in Week 3.

DATA SCIENCE AND VISUALIZATION - WEEK 3 REPORT

Spotify Dataset Cleaning ion and Outlier Detect

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team 9

Dataset: Spotify Tracks Dataset

1. Introduction

Week 3 focused on cleaning and preparing the Spotify dataset for further analysis. Data cleaning is an essential step in data science because missing values, duplicate records, and extreme observations can affect the reliability of analytical results.

In this stage, Python was used to identify and handle missing values, check duplicate records, and detect potential outliers using statistical methods and boxplots. The cleaned dataset was then verified and prepared for subsequent analysis.

2. Objectives

The main objectives of Week 3 were:

1. Identify missing values and handle them using suitable methods.
2. Check the dataset for duplicate records.
3. Detect potential outliers in numerical audio features.
4. Compare outlier detection results using statistical methods.
5. Verify and save the cleaned dataset for further analysis.

3. Tools and Technologies Used

The following tools and libraries were used:

- **Python:** Programming language for data cleaning and analysis.
- **Pandas:** Used to inspect, clean, summarize, and export the dataset.

- **NumPy:** Used for numerical operations.
- **Matplotlib:** Used to display graphs and visualizations.
- **Seaborn:** Used to generate boxplots.
- **Jupyter Notebook:** Used to execute Python code and display results.

4. Dataset Overview

The Spotify dataset contains information about tracks, albums, popularity, Spotify identifiers, and audio characteristics. The dataset was loaded and inspected using Pandas.

The notebook reported **19,335 rows and 46 columns**. Nine numerical audio features were selected for outlier detection:

1. Danceability
2. Energy
3. Loudness
4. Speechiness
5. Acousticness
6. Instrumentalness
7. Liveness
8. Valence
9. Tempo

These features were examined to understand their distributions and identify potential extreme observations.

5. Data Cleaning Methodology

5.1 Identification of Missing Values

Missing values were checked across the dataset by calculating the number of missing entries in each column.

The preview_url column contained 7,524 missing values, representing approximately 38.91% of the dataset. Several audio-related fields contained seven missing values each, representing approximately 0.04%.

This assessment helped identify columns that required attention during data cleaning.

5.2 Handling Missing Values

Missing values were handled using suitable imputation methods:

- **Numerical columns:** Missing values were filled using the median of the respective column.
- **Categorical or object columns:** Missing values were filled using the mode where a valid mode was available.
- **Verification:** The dataset was checked again after imputation to identify any remaining missing values.

The final missing-value check reported no remaining missing values.

5.3 Duplicate Record Detection

The dataset was checked for exact duplicate rows using Pandas. The drop_duplicates() method was applied to the working dataset.

The notebook reported zero duplicate rows. Therefore, the dataset remained at 19,335 rows and 46 columns after the duplicate-checking step

6. Outlier Detection and Visualization

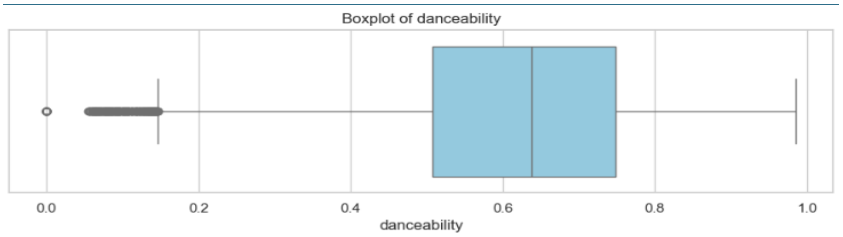
Outlier detection was performed to identify values that differed considerably from the general distribution of the selected audio features. Boxplots and statistical methods were used to identify potential outliers.

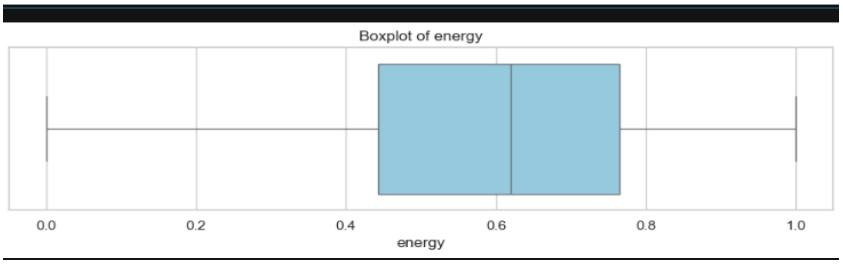
6.1 Boxplot Analysis

Boxplots were used to visualize the distributions of the nine numerical audio features. They display the median, interquartile range, and potential extreme observations.

The analysis identified potential outliers in several features, particularly instrumentalness, speechiness, loudness, and liveness. The presence of these values does not necessarily indicate incorrect data because extreme audio-feature values may represent genuine characteristics of songs.







6.2 Interquartile Range (IQR) Method

The IQR method is used to detect outliers by measuring the spread of the middle 50% of the data. It calculates the difference between the third quartile (Q3) and the first quartile (Q1).

IQR = Q3 - Q1

The method defines an acceptable range using the following limits:

- Lower limit: (Q1 - 1.5 times IQR)
- Upper limit: (Q3 + 1.5 \times IQR)

Any observation falling outside this range is identified as a potential outlier. This method was applied to the selected Spotify audio features to identify unusually low or high values.

6.3 Z-Score Method

The Z-score method is used to identify outliers by measuring how far an observed value deviates from the mean in terms of standard deviations.

[Z = frac{x-mu}{sigma}]

Where:

- (x) – Observed value
- (mu) – Mean of the dataset
- (sigma) – Standard deviation

In this study, values with an absolute Z-score greater than 3 ((|Z| > 3)) were identified as potential outliers. This method helped detect unusually high or low values in the selected Spotify audio features.

6.4 Comparison of Outlier Detection Results

The number of potential outliers identified by the IQR and Z-score methods is summarized below.

Audio Feature	IQR Outliers	IQR Percentage	Z-Score Outliers
Danceability	258	1.33%	90
Energy	0	0.00%	0
Loudness	1,813	9.38%	479
Speechiness	2,255	11.66%	336
Acousticness	0	0.00%	0
Instrumentalness	4,274	22.10%	886
Liveness	1,404	7.26%	524
Valence	0	0.00%	0
Tempo	70	0.36%	53

The IQR method identified the highest number of potential outliers in instrumentalness, followed by speechiness and loudness. The Z-score method also identified instrumentalness as the feature with the highest number of potential outliers, followed by liveness and loudness.

The difference in counts occurs because the two methods use different statistical criteria. These results indicate potential outliers rather than confirmed data errors.

6.5 Outlier-Handling Decision

Potential outliers were retained instead of being automatically removed. Extreme values in audio features may represent genuine differences between songs. Removing them without further investigation could result in the loss of useful information.

Therefore, outlier detection was used to identify observations for review rather than automatically delete them.

7. Descriptive Statistical Analysis

Descriptive statistics were examined to understand the central tendency and range of the selected audio features.

Audio Feature	Mean	Minimum	Maximum
Danceability	0.616	0.000	0.985
Energy	0.584	0.00002	1.000
Loudness	-8.605 dB	-54.376 dB	1.060 dB
Speechiness	0.107	0.000	0.954
Acousticness	0.309	0.000	0.996
Instrumentalness	0.102	0.000	1.000
Liveness	0.182	0.000	0.993
Valence	0.491	0.000	1.000
Tempo	119.795 BPM	0.000 BPM	230.401 BPM

The results show variation across the selected audio features. Danceability and energy had mean values of 0.616 and 0.584, respectively. Instrumentalness had a mean of 0.102, while the average tempo was approximately 119.795 BPM.

These statistics provide a numerical summary of the selected features and support further analysis of their relationship with song popularity.

8. Final Dataset Verification

After cleaning and outlier detection, the dataset was checked to confirm its final condition.

Verification	Result
Final number of rows	19,335
Final number of columns	46
Remaining missing values	0

Duplicate rows reported	0
Rows removed solely due to outlier flags	0
Exported file	spotify_cleaned.csv

The final checks indicated that the dataset contained no remaining missing values and no exact duplicate rows. Potential outliers were retained for further interpretation.

The cleaned dataset was exported as spotify_cleaned.csv for use in subsequent stages of the project.

9. Team Member Participation

The tasks for Week 3 were distributed among the team members to ensure systematic data preprocessing, analysis, and documentation.

S. No.	Team Member	Roll Number	Tasks and Responsibilities
1	Prithesa R	25BAD083	Dataset understanding, identification of missing values, and handling missing data using suitable imputation techniques.
2	Saravanan A	25BAD102	Duplicate record verification, data consistency checks, and final dataset validation after cleaning.
3	Tejas T	25BAD124	Outlier detection using the Interquartile Range (IQR) method and visualization of outliers using boxplots.
4	Viknesh T	25BAD128	Z-score-based outlier detection, descriptive statistical analysis, and preparation of the report documenting the results.

10. Challenges and Considerations

The following considerations were identified during the data-cleaning process:

1. The dataset contains metadata and URL fields that need careful handling.
2. The `preview_url` column contains many missing values.
3. IQR and Z-score methods identified different numbers of outliers.
4. Extreme values were retained, as they may represent valid musical characteristics. Further analysis is needed to study their relationship with song popularity.

11. Conclusion

Week 3 focused on cleaning and preparing the Spotify dataset for further analysis. Missing values were handled using median and mode imputation, duplicate records were checked, and potential outliers were investigated using boxplots, the IQR method, and Z-scores.

The final verification reported 19,335 rows and 46 columns, with no remaining missing values and no exact duplicate rows. Potential outliers were retained rather than automatically removed because extreme values may represent genuine musical characteristics.

The cleaned dataset was exported as spotify_cleaned.csv and is prepared for the next stage of the project: exploratory data analysis.